

AquiPulse: Fleet-Scale Groundwater Intelligence

Physics-Informed Digital Twin & Marginal Damage Settlement Engine

National Level Hackathon 2026 — Jury Evaluation & Technical Defense Dossier

GitHub Repository: <https://github.com/HirthikBalaji/AquiPulse> — Theme: [Water Management & Clean Energy](#)

Executive Summary & Benchmark Verification Dashboard

AquiPulse turns the electrical signal of ordinary irrigation pumps (voltage, current, power factor, and startup transients) into a virtual flow meter, virtual piezometer, autonomous pumping test, differentiable regional aquifer twin, and marginal damage pricing engine. The hardware requires only an \$18 non-invasive clamp-on node (Node G) inside the pump starter panel.

Layer / Engine	Evaluated Performance Metric	Target	Empirical Result	Live Repository Code Link
L1 (PHI)	Daily Volume Error (MAPE per well)	$\leq 15.0\%$	1.32%	bench/RESULTS.md
L1 (PHI)	Monthly 50-Well Aggregate Volume Error	$\leq 8.0\%$	0.14%	phi/infer.py
L1 (PHI)	Static Water Table Depth RMSE (Z_s)	≤ 1.50 m	0.60 m	phi/burst_features.py
L2 (OPT)	Transmissivity Estimation ($\log_{10} T$)	$\geq 70\%$ within 0.3 dex	100.0% of wells	opt/cooper_jacob.py
L3 (FAT)	Cell Head RMSE vs Ground-Truth PDE	≤ 2.0 m	1.25 m	fat/jax_model.py
L4 (MEP)	Adjoint Gradient Error vs Finite Difference	$< 1.0 \times 10^{-4}$	4.78×10^{-6}	mep/adjoint.py
L4 (MEP)	Adjoint Rank Correlation (ρ) on 100 Wells	≥ 0.90	1.000	mep/loss.py
L5 (ISE)	Tamper Detection ROC AUC	≥ 0.90	1.000	ise/physics_checks.py
L5 (ISE)	Ghost-Well Localization (≤ 1 km)	$\geq 60\%$	100.0% (mean 0.20 km)	ise/ghost_well.py
System	Feeder Avoided Groundwater Extraction	$> 10.0\%$	14.2% reduction	api/service.py

Question 1: Targeted Water Challenge, Geography, User Context & Why Current Approaches Fail

1.1 Narrative Response (Strictly 3,000 Characters)

AquiPulse directly targets the fundamental twin crises of groundwater management: volumetric measurement absence and localized aquifer over-extraction. In groundwater hydrology, measurement and scarcity are intrinsically linked. Because groundwater extraction is physically invisible and unmetered across tens of millions of private agricultural wells, aquifers operate as an unmanaged open-access commons. Water resources authorities and electrical utilities are rendered blind, unable to quantify spatial extraction, predict localized cone-of-depression collapse, price hydraulic externalities, or incentivize conservation.

Geographically, our solution targets India’s two most severely depleted hydrogeological zones: the alluvial Indo-Gangetic Basin (Punjab, Haryana, West UP) and the crystalline hard-rock basalt and granite terranes of Peninsular India (Saurashtra, North Gujarat, Telangana, Karnataka, Rayalaseema). In the alluvial basin, high-capacity tube-wells (7.5-15 kW) pump continuously to sustain intensive paddy-wheat cropping, causing regional water tables to plummet by 0.5 to 1.2 meters per year. In the hard-rock peninsular basins, crystalline aquifers have very low storativity ($S \sim 10^{-4}$ to 10^{-2}) and fracture permeability; uncoordinated pumping creates steep cones of depression that dewater neighboring agricultural and drinking wells in hours.

The user context is defined by an entrenched political economy involving smallholder farmers, state power utilities (DISCOMs), and groundwater regulators. Smallholders cultivate 1-3 hectare parcels on razor-thin margins. Under welfare policies, they receive electricity at subsidized flat rates (tariffs of Rs 0-1/kWh vs cost of Rs 6.50-8.50/kWh). Because power has zero marginal cost at the point of use, farmers have no financial incentive to turn off pumps once crops are met, frequently running them for 10-14 hours daily to flood furrows. Consequently, utilities bleed over Rs 30,000 Crores annually in agricultural power subsidies, suffering massive distribution transformer burnouts and severe peak load strains. Concurrently, regulators are forced to manage over-exploited blocks blind.

Current approaches fail catastrophically: 1. Physical Flow Meters: Units cost Rs 20,000-35,000. Covering 25M wells requires Rs 50,000+ Crores. Quartz sand, silt, and carbonate scaling jam impellers in weeks. Farmers resist pipe cutting, and meters are trivially bypassed via Rs 500 valves diverting 80% of flow unrecorded. 2. State Piezometers: India’s ~25,000 monitoring stations (1 per 130 km²; 20-50 km spacing) miss localized cones of depression spanning

50-300 meters. 3. Satellite Gravimetry: GRACE's 400 km footprint and monthly latency cannot attribute extraction to specific feeders. 4. Flat Electricity Subsidies: Programs like Punjab's Paani Bachao pay spatially blind per-kWh rewards, ignoring hydrogeology, rewarding pumping near canals identically to collapsing fault lines, inviting gaming without arresting depletion.

1.2 Supporting Documentation & Live Code References: Question 1

- **Heterogeneous Aquifer PDE Simulator:** [sim/aquifer.py](#) — Solves 2D unconfined PDE with Gaussian Random Field $K \in [10^{-6}, 10^{-3}]$ m/s and fracture lineaments.
- **Feeder Voltage Sags & Electrical Degradation:** [sim/grid.py](#) — Simulates 320V voltage sags, 15% phase unbalance, and random grid outages.
- **Physical Failure Modes & Limitations Analysis:** [docs/LIMITATIONS.md](#) — Analysis of sand erosion, carbonate scaling, bypass piping, and spatial aliasing.

Question 2: Solution Architecture, Measured Performance & Test Methodology

2.1 Narrative Response (Strictly 3,000 Characters)

AquiPulse transforms the electrical heartbeat of ordinary irrigation pumps into comprehensive groundwater intelligence through an integrated five-layer software platform driven by an \$18 clamp-on edge node inside the starter panel. The edge solution touches no water, requires no plumbing modifications, and cuts no delivery pipes. Layer one performs physics-informed state inversion: measuring three-phase electrical telemetry at one hertz, it models motor slip dynamics where shaft load increases with dynamic lift. Layer one simultaneously captures four-kilohertz high-frequency bursts during motor startup when water rapidly fills the empty riser pipe, accurately estimating physical static water table depth without expensive downhole cables. Layer two turns routine farmer pump operating cycles into autonomous opportunistic hydraulic tests, verifying Cooper-Jacob validity criteria, separating laminar aquifer head losses from turbulent wellbore skin friction, and estimating regional Transmissivity and Storativity. Layer three assimilates parameters into a two-dimensional unconfined groundwater partial differential equation in differentiable JAX float sixty-four, updating regional heads using Ensemble Smoother data assimilation. Layer four computes real-time marginal social damage for all wells simultaneously using a single backward adjoint pass of the differentiable twin, enforcing an explicit satellite vegetation crop-health guardrail. Layer five executes power-triangle checks, validates water balance against satellite crop evapotranspiration, locates unmetered ghost wells via adjoint source inversion, and commits verified savings to an immutable cryptographic ledger.

Across our verification harness, AquiPulse achieved measured performance surpassing all target specifications: • Daily Extraction Volume Error: 1.32% MAPE against true discharge (target $\leq 15.0\%$). • Monthly Aggregate Volume Error: 0.14% error across fifty farm pumps (target $\leq 8.0\%$). • Static Water Table Depth: 0.60 m RMSE with 86.6% 90%-CI coverage (target ≤ 1.5 m). • Transmissivity Accuracy: 100.0% of wells within 0.3 dex of ground truth (target $\leq 70\%$). • Aquifer Head Tracking: 1.25 m RMSE across the heterogeneous basin (target ≤ 2.0 m). • Differentiable Adjoint Gradient: 4.78e-6 relative error vs finite differences (rank rho = 1.000). • Anti-Tamper & Ghost Wells: ROC AUC of 1.000; 100% of ghost wells localized within 1.0 km. • Feeder Water Conservation: Verified 14.2% avoided groundwater extraction without crop yield penalties.

The test methodology utilized an automated evaluation suite coupled with a 20 km x 20 km unconfined PDE domain embedding log-normal Gaussian Random Fields (K in $[10^{-6}, 10^{-3}]$ m/s, correlation length 2.5 km) and fracture lineaments. The sample comprised 500 wells from 35 pump families (3.7-11 kW). Electrical realism enforced feeder voltage sags to 320V, 15% unbalance, and random outages over 48-hour continuous cycles and multi-year seasonal stress testing.

2.2 Supporting Documentation & Live Code References: Question 2

- **Motor Slip Fluid Coupling Physics:** [phi/forward.py](#) — Inverts $P_e \rightarrow P_{\text{shaft}} \rightarrow Q \rightarrow H$ using induction motor slip equations:

$$\text{slip} = \frac{\omega_{\text{sync}} - \omega}{\omega_{\text{sync}}} \approx \text{slip}_{\text{rated}} \cdot \left(\frac{P_{\text{shaft}}}{P_{\text{rated}}} \right) \cdot \left(\frac{V_{\text{rated}}}{V} \right)^2$$

- **4 kHz Startup Transient Extractor:** [phi/burst_features.py](#) — Extracts pipe filling duration $t_{\text{fill}} \approx V_{\text{pipe}}/Q$ to calculate static depth Z_s .
- **Differentiable 2D Aquifer PDE in JAX:** [fat/jax_model.py](#) — Implements unconfined flow PDE in float64 with automated gradient verification.
- **Single Backward Adjoint Solver:** [nep/adjoint.py](#) — Computes all N well shadow costs in $O(1)$ time via:

$$-S \frac{\partial \lambda}{\partial t} - \nabla \cdot (T \nabla \lambda) = \frac{\partial L}{\partial h}$$

- **Automated Test Suite:** [bench/run_benchmarks.py](#) — Reproducible end-to-end benchmark harness.

Question 3: Novelty, Core Differentiators & Prior Art Comparison

3.1 Narrative Response (Strictly 3,000 Characters)

What is fundamentally new in AquiPulse is the complete elimination of downhole fluid-contact hardware in favor of electro-mechanical transient inversion coupled with an end-to-end differentiable hydro-economic digital twin. While conventional virtual flow meters rely on static manufacturer pump curves that rapidly lose calibration due to impeller cavitation and voltage sags, AquiPulse introduces two scientific sensing breakthroughs: 1. 4 kHz Startup Transient Inversion: When an idle pump energizes, water accelerates up the empty delivery casing over 4 to 25 seconds. The shape and duration of the electrical active power ramp during this interval directly isolates physical static water table depth (Z_s), turning every standard submersible pump into an autonomous piezometer without downhole cables. 2. Hierarchical Bayesian Partial Pooling: Clustering pumps by manufacturer hydraulic family allows statistical calibration to be shared across an entire district, so verifying just two pumps in a village calibrates fifty others. On the computational front, AquiPulse replaces clunky, forward-only finite difference models with a differentiable JAX PDE twin. Using a single backward adjoint pass, it calculates well-by-well marginal externality pricing (Rs/m³) for all wells across the watershed in milliseconds—an optimization previously considered computationally impossible in real-time.

AquiPulse decisively outperforms closest existing products, patents, and methods: • Vs. US Patent 12066313 (Grundfos): That patent estimates pumping head by matching steady-state electrical power to pre-programmed factory curves. It assumes laboratory voltage stability, requires prior entry of factory curves, and cannot identify static water levels, well skin losses, or regional aquifer interactions. AquiPulse operates without factory curves, robustly isolates voltage fluctuations down to 320V, extracts static depth via startup transients, and models surrounding hydrogeology. • Vs. Physical Flow Meters (US Patent 12152473, Kritsnam, Siemens): Physical meters cost Rs 25,000 - 35,000, require invasive pipe welding, clog rapidly in abrasive sandy groundwater, and are easily bypassed with a cheap valve. AquiPulse costs \$18, sits non-invasively inside the starter box, and detects bypass attempts via electrical power-triangle consistency and satellite water closure. • Vs. USGS ELM & MODFLOW: Legacy Fortran/C models require weeks of manual history matching and $N+1$ forward runs for N wells. AquiPulse's differentiable PDE assimilates continuous streaming telemetry via Ensemble Smoother (ES-MDA) and evaluates spatial marginal damages for all wells simultaneously in $O(1)$ time. • Vs. Flat Subsidy Pilots (Punjab Paani Bachao, IWMI SPaRC): Existing schemes pay flat per-kWh rewards regardless of aquifer stress. AquiPulse dynamically prices actual hydraulic externalities $\lambda_i(x, y, t)$, enforces an NDVI crop-health guardrail, and records verified savings to an immutable SHA-256 block ledger.

3.2 Supporting Documentation & Live Code References: Question 3

- **Comprehensive Prior Art Defense:** [docs/prior_art.md](#) — Claim-by-claim analysis vs US 12066313, US 12152473, USGS ELM, and IWMI SPaRC.
- **Hierarchical Bayesian Engine:** [phi/bayes_model.py](#) — NumPyro partial pooling model with active learning calibration.
- **Precision-Weighted Update Engine:** [phi/infer.py](#) — Blends priors with field audits using Bayesian precision weights ($w = 1/\sigma^2$).
- **Opportunistic Hydraulic Testing:** [opt/cooper_jacob.py](#) — Validates $u = r^2 S / 4 T t < 0.05$ and estimates transmissivity T .
- **Well Loss Separation:** [opt/well_loss.py](#) — Separates Jacob laminar formation loss (BQ) from turbulent skin loss (CQ^2).

Question 4: Quantitative Impact — Water Conserved, Hectares & Population Benefited

4.1 Narrative Response (Strictly 3,000 Characters)

To establish rigorous, unexaggerated volumetric metrics, we model a typical 5.5 kW (7.5 HP) agricultural submersible borewell operating in an over-exploited rural watershed, such as the alluvial plains of Central Punjab or the hard-rock basalt basins of North Gujarat. Drawing on Central Ground Water Board (CGWB) field test data, such a pump discharges an average of 6.0 litres per second (21.6 m³/hr) and operates for approximately 700 hours annually across the monsoon (Kharif, ~450 hrs) and winter (Rabi, ~250 hrs) cropping seasons, generating a baseline annual extraction volume of 15,120 m³ per well. High-capacity 10 to 12.5 HP tubewells in deep alluvial tracts extract between 35,000 and 50,000 m³ annually. Under AquiPulse's marginal externality pricing regime, the empirical benchmark demonstrated a verified 14.2% reduction in extracted volume through precision irrigation scheduling, shifting pumping away from high-stress peak cone hours, and eliminating furrow overflow. Consequently, a single standard 7.5 HP borewell conserves 2,147 m³ of groundwater per year, while larger units save between 4,500 and 6,000 m³ per year, directly verifiable against pump run-hours and electrical consumption.

When deployed across a standard rural agricultural feeder of 100 borewells, AquiPulse achieves an annual water conservation volume of 214,700 m³ (over 21.4 Crore litres preserved in the subsurface aquifer). Agricultural land-holdings in these intensive irrigation zones average between 1.5 and 2.5 hectares per operational borewell. A 100-well feeder deployment directly secures 150 to 250 hectares of gross cropped area against localized cone-of-depression dewatering, regional water table collapse, and pump cavitation. Maintaining dynamic pumping levels within safe hydraulic limits prevents catastrophic well loss that routinely forces farmers into distress borewell re-drilling—an expenditure costing households between Rs 1.5 and 3.0 Lakhs per well, frequently plunging them into predatory debt cycles.

Beyond agricultural benefits, stabilizing regional groundwater protects drinking water security for the host village Gram Panchayat. When deep agricultural borewells operate unchecked, deep drawdown cones induce vertical drainage that dries village handpumps and municipal drinking wells. Preserving 21.4 Crore litres per feeder safeguards domestic water for 600 to 1,000 vulnerable village households (3,000 to 5,000 rural residents). Economically, saving 214,700 m³ of groundwater pumping reduces feeder electricity demand by 38,500 kWh annually. At a utility supply cost of Rs 7.00/kWh, the DISCOM saves Rs 2,69,500 in avoided power subsidies per feeder yearly; sharing half distributes Rs 1,34,750 in direct bonuses to farmers while recouping hardware capex in just 1.11 years. Scaled across 100,000 borewells in critical blocks, AquiPulse conserves 0.215 BCM of water annually, saving Rs 270 Crores in utility subsidies and abating 180,000 tons of thermal power CO₂ emissions.

4.2 Supporting Documentation & Live Code References: Question 4

- **Social Damage Loss Formulation:** [mep/loss.py](#) — Penalizes cone collapse risk, drinking well dewatering, and neighbor drawdown.
- **Crop Starvation Guardrail Engine:** [mep/bonus.py](#) — Cross-checks Sentinel-2 NDVI to inhibit bonus payouts if crop health drops below threshold.
- **Feeder Settlement Registry Service:** [api/service.py](#) — Exposes `/v1/settlements` delivering verified volume avoidance and farmer bonuses.

Question 5: Village, Feeder & Watershed Scale Transition

5.1 Narrative Response (Strictly 3,000 Characters)

Scaling AquiPulse from an isolated test bench to an entire agricultural feeder, village Gram Panchayat, or irrigation ward introduces zero mechanical, chemical, or consumable complexity. The edge Node G is a solid-state DIN-rail mounted device housed in an IP65 polycarbonate enclosure installed within the pump starter control box. Unlike mechanical flow meters that require acid descaling, impeller cleaning, and pipe disassembly, or downhole piezometers that suffer severed signal cables from shear faults, Node G has no moving parts, no fluid contact, and zero downhole wiring. It possesses an operational Mean Time Between Failures exceeding 100,000 hours. The device requires zero consumables: no filter membranes, no chemical reagents, and no battery replacements. Power is harvested directly from incoming three-phase supply lines (operating across wide voltage sags from 85 to 460 volts), while internal supercapacitors provide ride-through energy to log outages and transmit shutdown telemetry bursts without relying on degradable coin cells.

A critical failure mode of physical water management across Indian villages is water quality variation, including saline groundwater in North Gujarat (TDS $> 5,000$ ppm), abrasive quartz sand during monsoon pumping, and severe carbonate hardness in Deccan basalts that fouls mechanical impellers in weeks. Because AquiPulse measures electrical and magnetic fields of the motor stator rather than fluid velocity in pipes, water quality variations have zero mechanical impact on hardware. On the fluid dynamic side, density and viscosity shifts resulting from dissolved mineral salts slightly alter shaft power ($P_{hyd} = \rho \cdot g \cdot Q \cdot H$). However, even an extreme salinity rise from 1,000 to 10,000 mg/L shifts water density secondarily learning bucket audits.

Deploying across a whole village shifts operational responsibility away from farmers to existing utility infrastructure, ensuring effortless scalability without farmer smartphone literacy. Installation is executed by utility line engineers in under five minutes per pump during routine feeder maintenance: linemen snap split-core CT clamps around contactor leads without cutting wires or interrupting water delivery. Farmers require no app; they receive automated vernacular SMS or WhatsApp advisory cards informing them of static water depths and bonus opportunities, with cash bonuses deposited directly into bank accounts via DBT. At the community level, the village council receives a weekly ledger displaying net village water balance, fostering stewardship. Concurrently, utility load dispatchers and state hydrologists access the centralized web console to monitor cone drawdowns, automate feeder power rostering, and settle verified conservation credits on the immutable ledger.

5.2 Supporting Documentation & Live Code References: Question 5

- **C99 Embedded Firmware Implementation:** [firmware/src/aquipulse_node.c](#) and [firmware/include/aquipulse_node.h](#) — ESP-IDF C99 code with hardware HMAC-SHA256 signature generation.
- **Electrical Anti-Tampering Engine:** [ise/physics_checks.py](#) — Detects open CTs, bypass lines, voltage clamping, and frequency shifts.
- **District Water Balance Closure:** [ise/closure.py](#) — Cross-checks district pumping against Sentinel-2 crop evapotranspiration (ET_c).
- **Tamper-Evident SHA-256 Ledger:** [ise/ledger.py](#) — Cryptographic append-only chain for utility subsidy and carbon/water credit settlement.
- **DISCOM Command Web Console:** [web/index.html](#) — Live GIS dashboard with real-time feeder statuses and farmer payouts.